npj | science of learning

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Published in partnership with The University of Queensland

https://doi.org/10.1038/s41539-025-00369-4

Personalized and gamified auditory-cognitive training improves naturalistic speech-in-noise comprehension in older adults with hearing loss

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This study examined whether a gamified and personalized auditory-cognitive training (ACT) program could improve naturalistic speech-in-noise (SIN) comprehension in older adults with mild-to-moderate hearing loss. In a randomized controlled trial, 54 older participants with hearing loss were assigned to four weeks of ACT or an active control condition. SIN comprehension was assessed using conversational sentences embedded in cafeteria noise. Complementary measures assessed working memory, selective attention, phonological short-term memory, divided attention, speech intelligibility, subjective hearing ratings, and subjective listening effort. Participants completing ACT demonstrated significant improvements in SIN comprehension, with partial cognitive gains, which remained at follow-up. Active controls showed no improvements. SIN intelligibility did not change in either group, indicating a dissociation between low-level fidelity and higher-order comprehension. No changes emerged in subjective hearing reports. These findings underline ACT's potential for supporting SIN comprehension in older adults with hearing loss, offering a promising complement to traditional auditory rehabilitation.

Hearing loss (HL) is the most prevalent sensory deficit in older adults[1]. HL contributes to oral communication difficulties[2], social isolation[3], depression[4,5], cognitive decline[6], and dementia[7–9]. In addition, the prevalence of HL is projected to increase drastically within the next three decades (World Health Organization[10]), likely due to the aging trend in global population demographics and increasing recreational noise-exposure. Currently, the most common treatment of HL is sound amplification through hearing aids (Chisolm et al.[11]; Ferguson & Henshaw[12]). However, while sound amplification improves sound audibility[13], it leaves other common issues amongst HL-affected older adults unaddressed[13–15].

Specifically, regardless of whether they use hearing aids—including those with modern noise reduction algorithms—many older adults with HL continue to report difficulties understanding speech in the presence of competing background noise[16,17]—often referred to as speech-in-noise (SIN) comprehension. Older adults' need for SIN comprehension is especially high during social settings in noisy environments[18] such as restaurants or family gatherings. In case of inability to meet these SIN comprehension demands, a reduction in social engagement and potential social isolation follows[3]. Like HL, social isolation is further associated with worse outcomes for cognitive decline and dementia, and may in some cases mediate this relationship[19,20]. Accordingly, interventions addressing the cognitive deficits behind SIN comprehension difficulties in older adults with HL are urgently needed.

A prominent example of a speech-in-noise situation is the exhaustively investigated cocktail party problem[18,21], typically referring to the demanding task of focusing on a single conversation in the presence of other competing auditory inputs[22] such as other speakers or environmental noises. Results from studies on SIN comprehension in cocktail party scenarios gave rise to the important notion that, alongside sensory hearing capacity, cognitive ability plays an essential role in SIN comprehension performance[23–26]. Specifically, SIN comprehension is proposed to depend on phonological short-term memory[27], attention[28–30], and especially working memory[25,31–34]. These same cognitive components are likewise deemed essential in contemporary models of speech/language processing[35,36]. Notably, the contribution of working memory to SIN comprehension seems to increase with age and the individual degree of HL[37,38].

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Unfortunately, even modern sound amplification and noise reduction algorithms merely address the sensory (bottom-up) components of SIN comprehension and cannot account for the cognitive (top-down) deficits that typically occur with aging. As a result, current hearing aids appear unable to completely resolve SIN comprehension difficulties in older adults with HL[39–43]. There exists a clear necessity for complementary interventions directly addressing cognitive deficits in recipients of auditory rehabilitation[44].

One promising approach for addressing this need and improving top-down processing during SIN comprehension is the principle of auditory-cognitive training (ACT). As the name suggests, ACT commonly aims to improve auditory perceptual and cognitive abilities in tandem. On an auditory perceptual level, individuals train to improve sensory bottom-up processing of acoustic stimuli[45,46]. From a cognitive perspective, ACT protocols further aim to specifically improve cognitive abilities that are thought to support acoustic noise inhibition, speech comprehension, and detection[47]. There is a positive consensus regarding the potential of ACT for auditory rehabilitation[44], although results on ACT efficacy are by no means unanimous[44]. Such variability is likely due to a range of methodological shortcomings in studies assessing ACT efficacy. For one, studies investigating ACT tend to either lack appropriate, active control groups[48] or forego control groups altogether[44]. Furthermore, investigation of ACT efficacy is mostly pursued without incorporating naturalistic outcome measures, instead relying on highly structured laboratory tasks, making it difficult to evaluate the transferability of ACT benefits to real-life listening situations[49]. Moreover, studies often omit untrained outcome tasks using novel stimuli, limiting conclusions about transfer beyond the trained content. Such tasks are essential to assess generalization to real-world listening, which typically differs from the training context[50].

Addressing these shortcomings, the present study investigates the efficacy of gamified personalized ACT on improving speech comprehension under naturalistic listening and response conditions, in a sample of hearing-impaired older adults with and without hearing aids. To further examine whether potential cognitive improvements measured by the ACT training app tasks generalize to independent assessments, the protocol includes a laboratory-based test battery covering the same four domains targeted by the ACT (phonological short-term memory, working memory, divided attention, selective attention).

Following the presented line of argumentation, we hypothesize that H1: Participants undergoing auditory-cognitive training show greater improvements in naturalistic speech-in-noise comprehension compared to those in an active control group. Likewise, we expect that H2: Auditory-cognitive training results in improved performance on tasks for each targeted cognitive domain (working memory, selective attention, divided attention, phonological short-term memory).

Results

Baseline group comparisons

Welch's independent samples t-tests were conducted to assess group equivalence between the experimental and active control groups on demographic, training-related and cognitive characteristics. No significant baseline differences were observed between groups in participant age ($t(31.48) = 1.76$, $p = 0.177$), PTA ($t(39.68) = -0.87$, $p = 0.775$), MoCA scores ($t(37.36) = -0.91$, $p = 0.735$), SSQ ($t(44.97) = 0.12$, $p = 1$), or training intensity ($t(42.96) = -0.31$, $p = 1$). Likewise, no significant group differences were evident in phonological short-term memory ($t(33.24) = -1.39$, $p = 0.348$), verbal working memory ($t(47.18) = -0.70$, $p = 0.977$), selective attention ($t(36.69) = -0.19$, $p = 1$), or divided attention ($t(38.78) = -1.89$, $p = 0.132$). Descriptive statistics for all demographic and cognitive comparisons are provided in Supplementary Table 1 and illustrated in Fig. 1.

On-training cognitive tasks

Significant improvements in the experimental group's on-training cognitive performance from pre- to post-training were observed across all four cognitive domains. Working memory performance increased robustly over time ($\beta = 27.92$, 95% CI [14.89, 40.93], $p < 0.001$, $d = 2.77$). Likewise, phonological short-term memory showed a significant improvement following training ($\beta = 12.18$, 95% CI [1.89, 22.47], $p = 0.020$, $d = 1.21$). In the domain of selective attention, participants demonstrated a reliable gain in performance from pre- to post-training ($\beta = 11.20$, 95% CI [5.67, 16.73], $p < 0.001$, $d = 1.08$). Similarly, divided attention improved significantly over time as well ($\beta = 18.51$, 95% CI [9.22, 27.81], $p < 0.001$, $d = 1.03$).

As for further covariates, no significant effects of PTA were observed in any cognitive domain. Similarly, age did not show consistent associations, though a marginally negative trend was seen for selective attention ($\beta = -2.53$, 95% CI [−5.32, 0.25], $p = 0.074$). Notably, hearing aid use showed significant positive associations with performance, specifically in phonological short-term memory ($\beta = 17.58$, 95% CI [2.13, 33.03], $p = 0.026$) and divided attention ($\beta = 17.02$, 95% CI [1.72, 32.31], $p = 0.027$). A summary of model results is provided in Supplementary Table 2, with effects being illustrated in Fig. 2.

Off-training cognitive tasks

Concerning lab-conducted off-training cognitive tasks, done by both groups, significant across-group performance improvements from pre- to post-training were observed for phonological short-term memory (PSTM), working memory (WM; $\beta = 8.92$, 95% CI [3.57, 14.27], $p = 0.001$, $d = 1.07$), focused attention (FA; $\beta = 14.34$, 95% CI [5.94, 22.74], $p = 0.001$, $d = 1.01$), but not divided attention (DA; $\beta = 1.50$, 95% CI [−7.00, 9.99], $p = 0.729$, $d = 0.08$). Across all cognitive domains, no significant group session interactions were detected, indicating improvements were not statistically specific to the auditory-cognitive training condition.

Post-hoc simple effects analyses indicated that PSTM improvements were significant in both the control (CG-HL: $\beta = 11.64$, 95% CI [0.23, 23.00], $p = 0.048$, $d = 0.34$) and experimental group (EG-HL: $\beta = 12.26$, 95% CI [2.91, 21.61], $p = 0.011$, $d = 0.44$). For WM and FA, significant gains were observed only in the experimental group (WM: $\beta = 10.73$, 95% CI [4.54, 16.91], $p = 0.001$, $d = 0.58$; FA: $\beta = 15.27$, 95% CI [5.89, 24.65], $p = 0.002$, $d = 0.55$), whereas the control group showed no significant change for WM ($\beta = 7.12$, 95% CI [−1.67, 15.91], $p = 0.114$, $d = 0.27$) and only a non-significant trend for FA ($\beta = 13.40$, 95% CI [−0.58, 27.37], $p = 0.064$, $d = 0.32$). No significant pre- to post-training changes were observed for DA in either group (CG-HL: $\beta = 2.59$, 95% CI [−11.00, 16.18], $p = 0.710$, $d = 0.06$; EG-HL: $\beta = 0.40$, 95% CI [−9.74, 10.53], $p = 0.938$, $d = 0.01$).

Neither pure-tone average (PTA), hearing aid use, nor age showed consistent effects across cognitive domains. Training trajectories per cognitive domain are displayed in Fig. 3. A comprehensive overview of model estimates is provided in Supplementary Table 3.

Sustain of cognitive training effects

At follow-up, the performance gains observed immediately after training were maintained or further enhanced across cognitive domains. To examine these patterns, simple effects analyses were conducted within each group across sessions. For phonological short-term memory, both the control group (CG-HL) and the experimental group (EG-HL) demonstrated significant improvements from pre-training to post-training (CG-HL: $\beta = 14.59$, 95% CI [1.67, 27.51], $p = .028$, $d = 1.48$; EG-HL: $\beta = 10.40$, 95% CI [1.01, 19.80], $p = 0.030$, $d = 1.05$). These gains were sustained at follow-up, with scores remaining significantly higher than baseline in both groups (CG-HL: $\beta = 13.83$, 95% CI [0.91, 26.75], $p = 0.037$, $d = 1.40$; EG-HL: $\beta = 13.54$, 95% CI [4.14, 22.94], $p = 0.005$, $d = 1.37$). No significant declines were observed between post-training and follow-up (both ps >0.50), suggesting durable training effects. Selective attention also showed significant post-training improvements (CG-HL: $\beta = 17.28$, 95% CI [1.70, 32.87], $p = 0.030$, $d = 1.75$; EG-HL: $\beta = 13.34$, 95% CI [4.76, 21.92], $p = 0.003$, $d = 1.35$), with performance remaining significantly elevated at follow-up relative to baseline (CG-HL: $\beta = 21.19$, 95% CI [5.61, 36.78], $p = 0.008$, $d = 2.14$; EG-HL: $\beta = 23.74$, 95% CI [14.76, 32.72], $p < 0.001$, $d = 2.40$). In the experimental group, scores continued to improve between post-training and follow-up ($\beta = 10.40$, 95% CI [1.46, 19.35], $p = 0.019$, $d = 1.05$), indicating ongoing benefits beyond the immediate post-training period. For

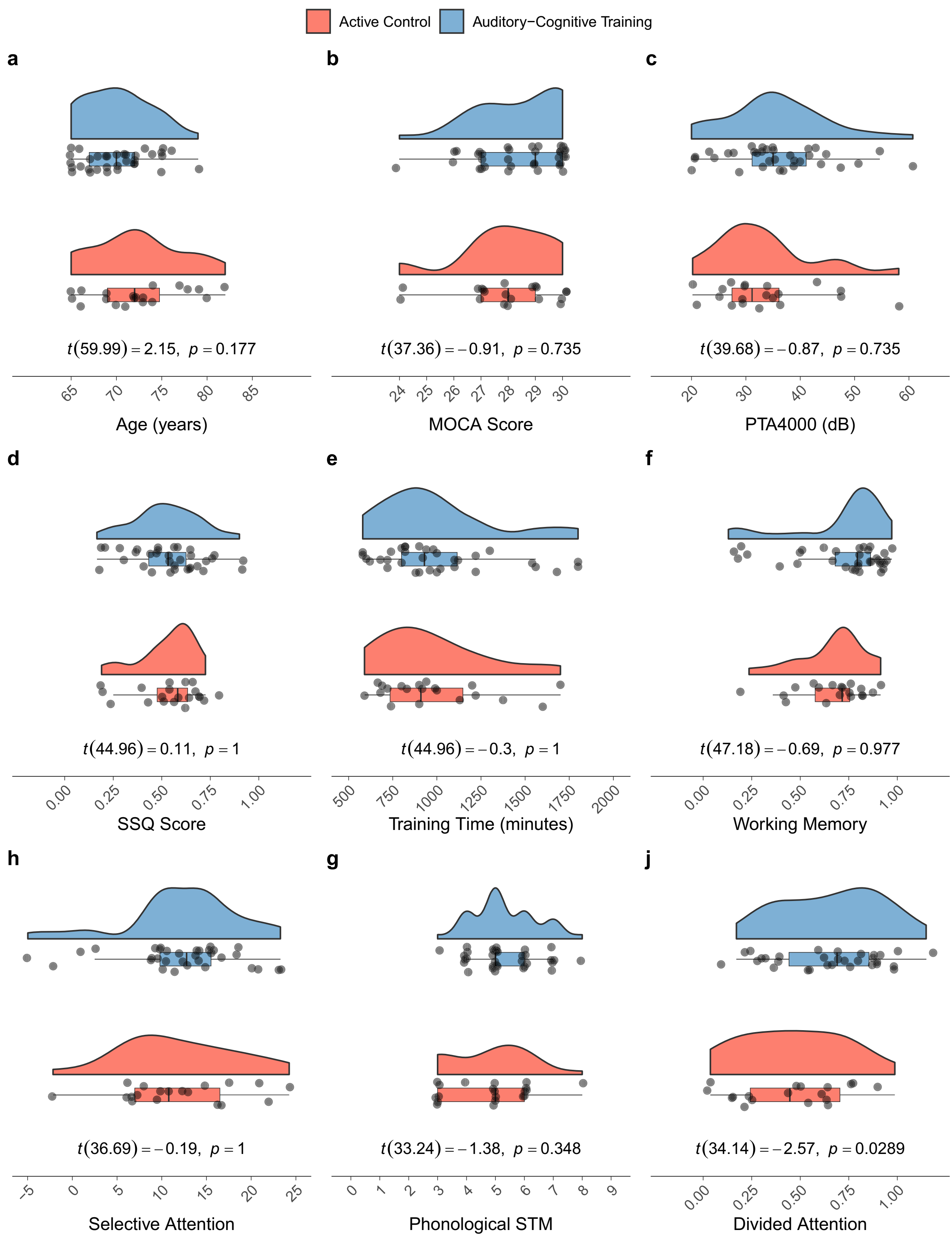

Fig. 1 | Baseline raw value distribution comparisons across groups. Note: Shown are raincloud plots illustrating baseline group comparisons for age (**a**), general cognitive performance (MoCA score; (**b**), hearing thresholds (PTA4000; (**c**), subjective hearing ability (SSQ score; (**d**), and total training duration in minutes (**e**), as well as cognitive performance across four domains: working memory (**f**), selective attention (**h**), phonological short-term memory (**g**), and divided attention (**j**). Data is shown for the active control group (red) and the experimental group (blue). Individual data points are represented by dots, group distributions by half-density curves, and means by horizontal bars. *t*-values and uncorrected p-values from independent-samples t-tests are displayed beneath each panel. No group differences in baseline measures were found.

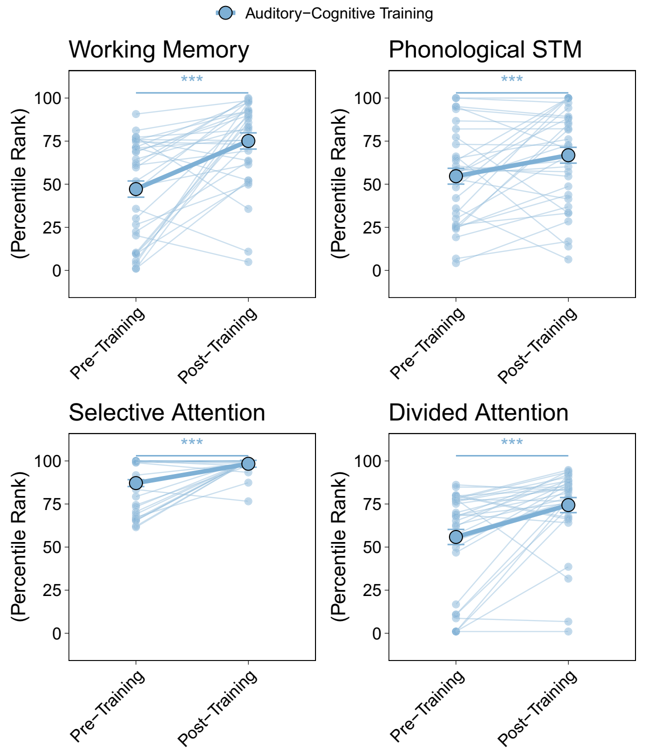

Fig. 2 | Trajectories of training intervention cognitive tasks for phonological short-term memory, working memory, selective attention, and divided attention. Note: Line plots display significant improvements from pre-training to post-training on the trained cognitive tasks: Working Memory, Phonological Short-Term Memory, Selective Attention, and Divided Attention. Error bars represent standard errors of the estimated marginal means.

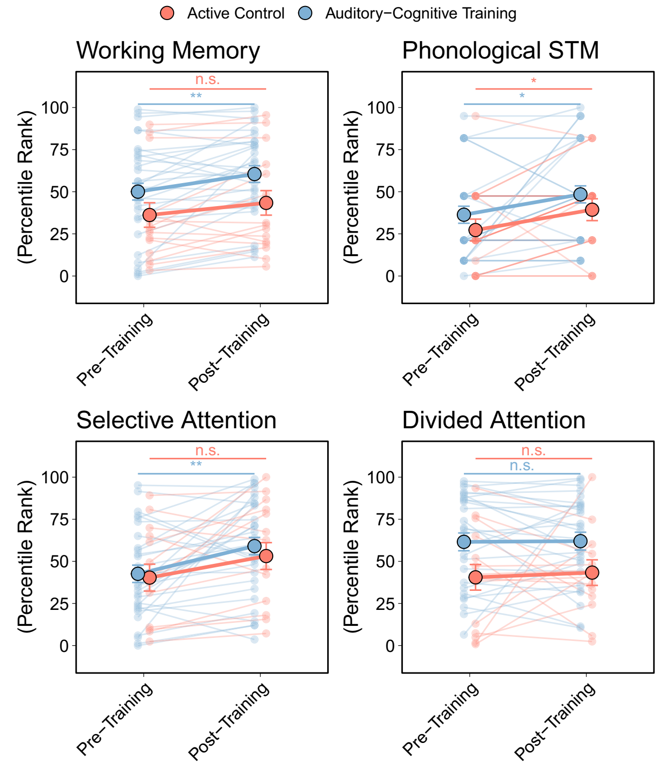

Fig. 3 | Trajectories of untrained cognitive tasks for phonological short-term memory, working memory, selective attention, and divided attention. *Note.* Estimated marginal means of performance on lab-based off-training cognitive tasks measured at pre-training, post-training, and follow-up sessions, separately for the experimental group (blue) and active control group (red). Shown are scores for Phonological Short-Term Memory (STM), Working Memory, Selective Attention, and Divided Attention. Significance indication is based on post-hoc simple effect analysis. Error bars represent standard errors.

working memory, significant gains from pre-training to post-training were found in the experimental group ($\beta = 10.68$, 95% CI [2.18, 19.18], $p = 0.014$, $d = 1.08$). At follow-up, performance remained higher than baseline, although the comparison approached but did not reach conventional significance levels ($\beta = 7.99$, 95% CI [− 0.47, 16.44], $p = 0.064$, $d = 0.81$). No significant differences were detected between post-training and follow-up, suggesting maintenance of the initial improvements. Divided attention showed no significant changes across sessions in either group.

Speech-in-noise comprehension after auditory-cognitive training

Employing GLMMs on single-trial accuracy data, we identified a significant interaction effect between training session and group (OR = 1.47, 95% CI [1.25, 1.72], $p < 0.001$), demonstrating differential improvements in speech-in-noise (SIN) comprehension from pre- to post-training across groups. Specifically, the experimental group exhibited significantly greater improvements in post-training performance compared to the active control group. The main effect of session alone was not significant (OR = 1.02, 95% CI [0.81, 1.27], $p = 0.897$).

Significant main effects were observed for noise level, with comprehension accuracy differing significantly between conditions. Compared to the reference level (no-noise condition), SIN comprehension accuracy decreased significantly under the low-noise condition (OR = 0.06, 95% CI [0.03, 0.12], $p < 0.001$) and showed an even larger decrease under the high-noise condition (OR = 0.006, 95% CI [0.003, 0.013], $p < 0.001$). Additionally, higher pure-tone averages (pta_{4000}) significantly predicted reduced comprehension accuracy (OR = 0.65, 95% CI [0.56, 0.76], $p < 0.001$). Age showed a trend towards significance, suggesting slightly lower comprehension accuracy with increasing age (OR = 0.85, 95% CI [0.71, 1.02], $p = 0.079$). Hearing aid use did not significantly impact comprehension accuracy (OR = 0.93, 95% CI [0.65, 1.32], $p = 0.681$).

Significant interaction effects were found between session and noise level, indicating that training-related gains in SIN comprehension were smaller under both low-noise (OR = 0.77, 95% CI [0.60, 0.98], $p = 0.035$) and high-noise conditions (OR = 0.82, 95% CI [0.65, 1.09], $p = 0.104$), relative to the no-noise condition. Furthermore, a significant three-way interaction between session, group, and noise level showed training-related improvements in the experimental group to be moderated by noise condition, with attenuated gains observed under high-noise (OR = 0.77, 95% CI [0.67, 0.89], $p < 0.001$) and low-noise conditions (OR = 0.84, 95% CI [0.73, 0.96], $p = 0.009$) compared to no-noise trials.

A considerable portion of the overall variance was attributable to individual differences between participants and variability across sentence items, as indicated by the random-effects structure–highlighting the importance of modeling item-level variability when using ecologically valid, naturalistic speech materials. Effects are visualized in Fig. 4. Full model output, including odds ratios, confidence intervals, and random effects estimates, is provided in Supplementary Table 4.

Effect of PTA on SIN accuracy in different noise levels

To further investigate how the significant main effect of hearing loss affected speech-in-noise (SIN) comprehension across noise conditions, we ran a post hoc model including an interaction between pure-tone hearing (PTA) thresholds and noise level. Results confirmed a robust main effect of PTA, indicating that greater hearing loss was associated with reduced comprehension accuracy overall (*OR* = 0.62, 95% CI [0.43, 0.90], $p = 0.011$).

Figure 5 illustrates a consistent pattern across groups, with comprehension declining alongside increasing PTA, and the effect becoming more pronounced under higher noise levels. This pattern underscores the

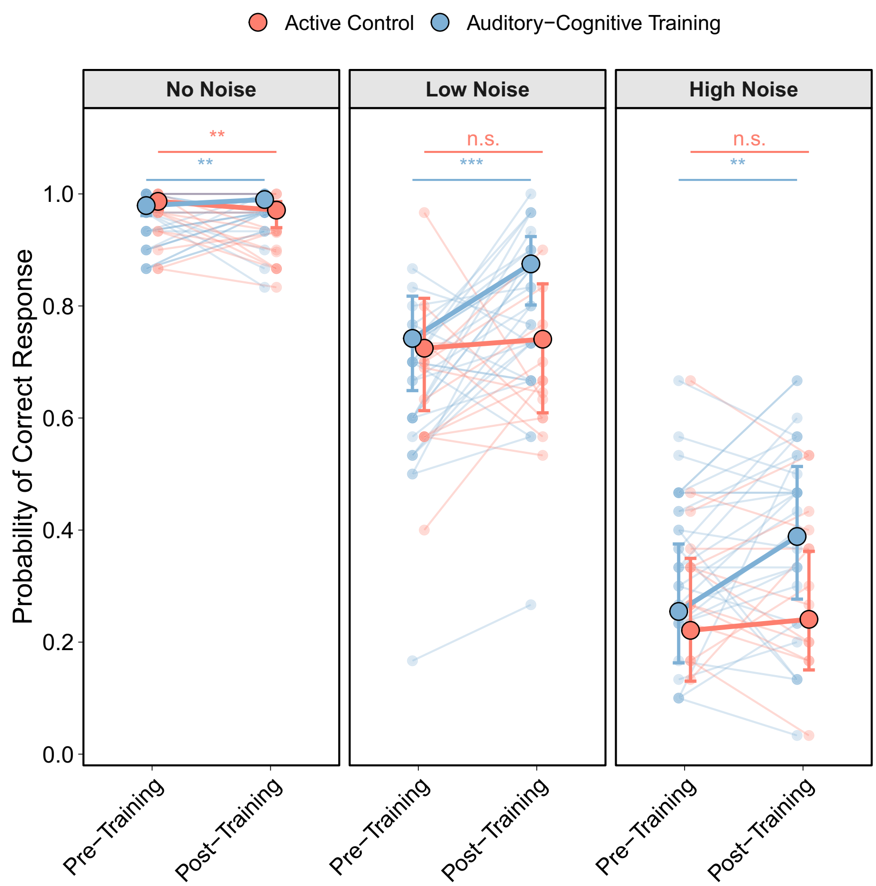

Fig. 4 | Training effect on speech-in-noise comprehension. Note: Displayed are estimated marginal means of speech-in-noise comprehension accuracy (probability of correct response) measured at pre-training and post-training sessions for each noise condition (No Noise, Low Noise, High Noise), separately for participants in the experimental group (blue) and active control group (red). Error bars represent standard errors.

cumulative challenge posed by combined auditory degradation and adverse listening environments.

The inclusion of the PTA noise level interaction did not alter the pattern of other main findings, including the significant interaction between session and group ($OR = 2.10$, 95% CI [1.32, 3.32], $p = 0.002$), which again confirmed training-related improvements in the experimental group. Age remained a significant predictor ($OR = 0.83$, 95% CI [0.69, 0.99], $p = 0.040$), while hearing aid use and divided attention scores were not significantly associated with comprehension performance.

Cognitive predictors for auditory-cognitive training benefit

To examine whether baseline cognitive abilities predicted training-related improvements in speech-in-noise (SIN) comprehension, a multiple regression model was fitted within the experimental group (EG-HL). Phonological short-term memory (PSTM), working memory (WM), focused attention (FA), divided attention (DA), age, hearing thresholds (PTA), and hearing aid use were entered as predictors. No significant associations were observed for PSTM ($\beta = 0.0005$, 95% CI [− 0.0004, 0.0013], $p = 0.294$), WM ($\beta = -0.0005$, 95% CI [− 0.0015, 0.0005], $p = 0.324$), FA ($\beta = -0.0005$, 95% CI [− 0.0013, 0.0003], $p = 0.178$), DA ($\beta = 0.0005$, 95% CI [− 0.0002, 0.0012], $p = 0.166$), PTA_{4000} ($\beta = 0.0061$, 95% CI [− 0.0166, 0.0289], $p = 0.593$), or hearing aid use ($\beta = 0.0200$, 95% CI [− 0.0223, 0.0622], $p = 0.349$). Age was marginally associated with SIN improvement ($\beta = 0.0234$, 95% CI [0.0001, 0.0466], $p = 0.050$).

Subjective hearing, listening effort, or speech-in-noise intelligibility

The results from generalized linear mixed models (GLMMs) indicated no significant effects of ACT on subjective hearing capacity, listening effort, or speech-in-noise intelligibility. Subjective hearing capacity, as per SSQ-12, showed no significant changes across sessions or between groups. The interaction between measurement times and group assignment was also not significant, indicating no treatment-specific improvement of subjective hearing capacity. Notably, higher PTAs were significantly associated with OLSA ($OR = 1.21$, 95% CI [0.78, 1.64], $p < 0.001$). Pre-to post-training trajectories for SSQ, LE, and OLSA are displayed in Fig. 6. A comprehensive summary of model estimates is provided in Supplementary Table 5.

Discussion

Hearing loss (HL) is a common sensory impairment among older adults, and if untreated, associated with cognitive decline, social withdrawal, and reduced quality of life. While hearing aids enhance audibility, many individuals continue to struggle with understanding speech in noisy environments, one of the most frequently reported challenges among the hearing-impaired. This randomized controlled trial examined whether a four-week, gamified, personalized, and commercially distributable auditory-cognitive training (ACT) program could improve naturalistic speech-in-noise (SIN) comprehension in older adults with mild-to-moderate HL.

Relative to an active control group completing foreign language training, the experimental group undergoing ACT showed significant improvements in SIN comprehension accuracy. Beyond SIN comprehension, participants in the experimental group further demonstrated significant improvements across several cognitive domains, including working memory, selective attention, and phonological short-term memory—effects that were partially maintained at a 2-month follow-up. These cognitive gains confirm fundamental efficacy of gamified and personalized ACT for improving capacities across targeted cognitive domains, while the alignment between cognitive gains and comprehension benefits underscores the relevance of domain-general cognitive capacity in real-world listening. Notably, no improvement was observed for divided attention, possibly reflecting limited transfer for multitasking-related abilities within the timeframe of short-term training.

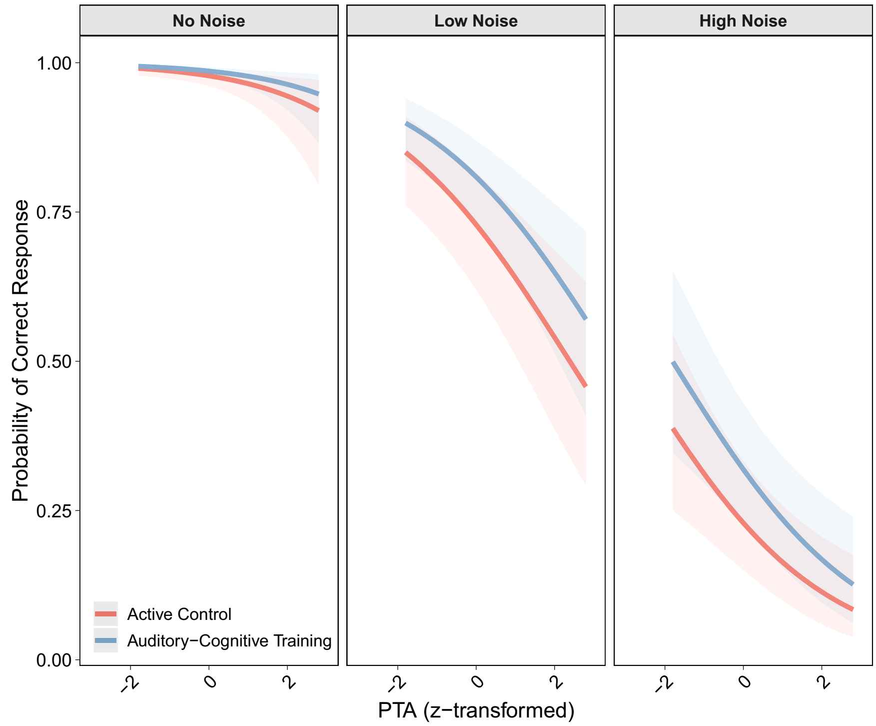

Fig. 5 | Predicted probability of correct speech-in-noise comprehension responses as a function of hearing loss (PTA) across noise levels. Note: Shown Displayed model estimates showing the probability of correct responses in the speech comprehension task plotted against z-standardized pure-tone average thresholds (PTA). Panels represent the three noise conditions (No Noise, Low Noise, High Noise). Shaded areas indicate 95% confidence intervals. Blue lines reflect participants in the experimental group, red lines those in the active control group. A stronger decline in performance with increasing PTA is observed amidst increasing noise levels.

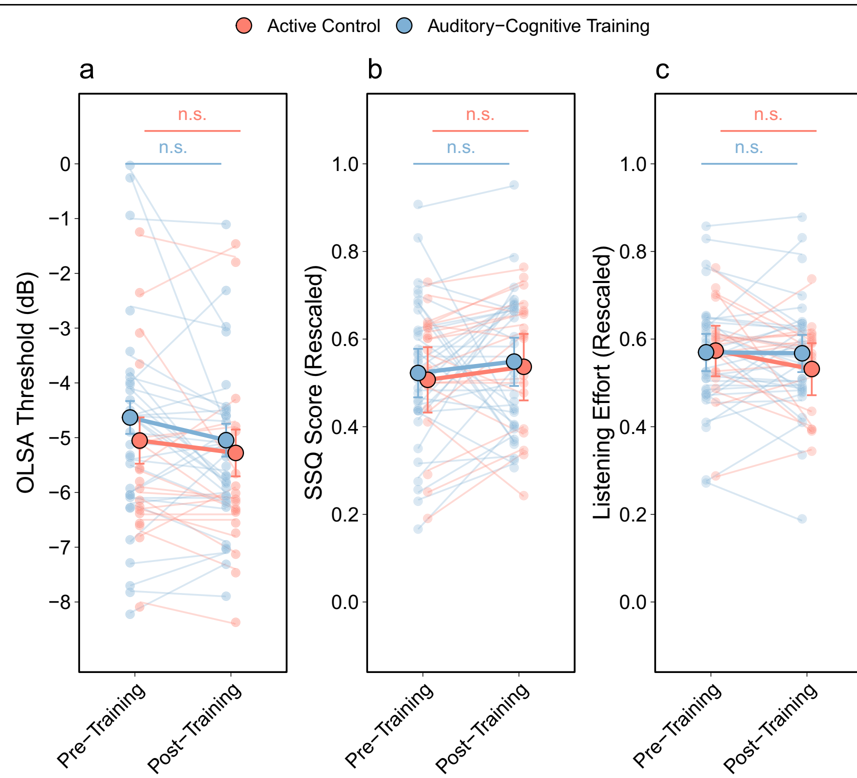

Fig. 6 | Training affects speech intelligibility, subjective hearing, and listening effort. Note. Line plots displaying estimated marginal means for speech intelligibility (OLSA threshold; (**a**)), subjective hearing ability (SSQ score; (**b**)), and perceived listening effort (**c**) at pre-training and post-training sessions, separately for participants in the experimental group (blue) and the active control group (red). Error bars indicate standard errors.

Interestingly, hearing aid users showed increased post-training performance benefit in certain cognitive domains. Although it poses a result of secondary interest, it may suggest that ACT is particularly effective when cognitive and sensory interventions are combined, warranting further investigation into the matter. Additionally, while employed training tasks were embedded in a gamified mobile interface, lab tasks followed classical paradigms on unfamiliar desktop devices. Thus, partial transfer of training benefit indicates partially successful generalization of training benefits beyond the immediate training environment.

Generally, these findings align with the Ease of Language Understanding (ELU) model[36], which posits that cognitive resources become increasingly critical as auditory input becomes degraded. Improvements in SIN comprehension occurred without corresponding changes in speech intelligibility thresholds (OLSA), highlighting a dissociation between low-level auditory fidelity and higher-order comprehension. This pattern is consistent with recent findings that emphasize the role of cognitive processes in challenging listening conditions[51]. Adding to this perspective, Fisher et al.,[52] recently demonstrated that short-term, home-based auditory-cognitive training not only enhances behavioral SIN comprehension, but also modulates neural speech tracking—further underscoring an involvement of cognitive and neural mechanisms in SIN comprehension difficulties.

Despite improvements in objective outcomes, subjective measures (SSQ-12, listening effort ratings) did not significantly change. This divergence between objective and subjective results has been observed previously (Ferguson et al.,[53]; Henshaw & Ferguson[45]) and may reflect limitations in metacognitive awareness or persistent perceived effort.

The present study advances the ACT literature by addressing critical methodological shortcomings in earlier work. Specifically, we incorporated an active control group to mitigate expectation effects, employed ecologically valid outcome measures, and evaluated both on- and off-training cognitive tasks to assess transfer. In contrast to prior work[44] that often lacked adequate controls or relied solely on lab tasks, our design offers stronger evidence for real-world applicability.

The gamified nature of ACT deserves particular attention. Gamification elements—such as adaptive difficulty, reward systems, and engaging visuals—are known to enhance motivation and adherence[54,55]. This is especially relevant for older adults, where sustained engagement remains a challenge. Previous research suggests that gamified formats not only improve compliance but may also enhance efficacy[56]. Our results further support the feasibility and effectiveness of gamified ACT in aging populations.

Another key innovation in this study was the individualized training approach. Personalized regimens based on baseline cognitive profiles are known to boost training efficiency and outcome magnitude[48]. Dynamic adaptation of difficulty levels and content in the employed ACT may have thus contributed to the strong training effects observed here.

Although ACT involved both cognitive and sensory components, the sensory aspects were integrated into broader cognitive tasks, supporting the conclusion that observed gains were primarily driven by cognitive improvements. This interpretation aligns with findings from other multimodal training paradigms, such as music interventions[57], which show benefits across cognitive and perceptual domains.

Nonetheless, several limitations must be acknowledged. First, SIN comprehension was not measured at follow-up, limiting conclusions about the long-term durability of observed effects. Additional limitations include the relatively modest sample size, reliance on self-reported training adherence in the control group, and the absence of item-level data in lab-based cognitive tasks. These constraints reduced our capacity to explore within-subject variability. Future research should extend ACT duration, refine personalization protocols, and integrate metacognitive supports to enhance awareness of improvements. Ecological momentary assessments may further provide richer data on real-world communication outcomes.

In conclusion, the present study provides novel evidence on the potential of gamified, personalized auditory-cognitive training to support improvements in naturalistic SIN comprehension in older adults with HL. By targeting cognitive mechanisms, such interventions offer a scalable and effective complement to traditional auditory rehabilitation like hearing aids, with the potential to improve daily communication and promote healthy aging.

Methods

Participants

The studied sample consisted of 60 right-handed German-speaking older adults ($M_{Age} = 71$, $Range_{Age} = 65–82$, $SD_{Age} = 4.53$; 30 male) with mild to moderate HL (20–60 dB PTA; Chadha et al.,[58]). Six participants were excluded from data analysis because they did not complete all three measurements, resulting in a total sample of 54 participants. Of these, 21 participants were regular hearing aid users ($N_{EG\text{-}Aided} = 16$; $N_{CG\text{-}Aided} = 5$), and had been using their devices for a minimum of 6 months prior to study participation. The sample was primarily recruited through public advertisements, public lectures on SIN comprehension, and the use of physical flyers.

A structured health questionnaire ensured that no participant had a history of developmental, neurological, or psychiatric disorders. No participants reported chronic tinnitus, simultaneous bilingualism, or asymmetric hearing loss (<15 dB interaural threshold differences). Likewise, professional musicians were excluded from study participation, due to musical experience accompanying significantly better frequency and tone discrimination[59]. All participants showed age-appropriate cognitive status ($M_{MoCA} = 28.2$, $Range_{MoCA} = 24–30$, $SD_{MoCA} = 1.6$), as measured by the MoCA (Montreal Cognitive Assessment ≥ 23; Carson et al., 2018; Nasreddine et al., 2005).

Verbal working memory

Verbal working memory was measured using a variant of the reading span task (RS), originally designed by Daneman & Carpenter[60]. This RS, alongside its parameters and German stimuli, was adapted from the validated working memory test battery of Lewandowsky and colleagues[61]. Participants were asked to read a given number of sentences, each being followed by a consonant. Sentences had to be judged as either sensical (e.g. "*Humans typically have a nose.*") or nonsensical (e.g. "*Bees are typically green.*"), using two dedicated arrow keys. The stimulus set contained an equal number of meaningful and meaningless sentences, most of them following the same "*X is Y*" structure. Presented consonants had to be memorized, with a question mark at the end of each trial cueing participants to enter all memorized consonants from the current set in correct forward order. Sentence response timeout occurred after 5000 ms. Consonants were presented for 1000 ms. The inter-stimulus interval was set to 1500 ms. List length of trials was defined as the number of sentence-consonant pairs that were presented within it. Three trials were presented per list length, in ascending order. List length across blocks varied from three to seven. Task onset was preceded by three practice trials of list lengths two, three, and four. Accuracy was calculated using partial credit scoring. As such, accuracy scores represented the proportion of correctly recalled letters across all non-practice trials.

Phonological short-term memory

A forward digit span (DS; Woods et al.,[62]) was employed for testing phonological short-term memory. Digits were randomly generated for each measurement, using three generative constraints. Specifically, no sequence could begin or end with a zero or one, adjacent digits could not differ by 1, and a sequence could not contain the same digit twice (Wolfe, 2020). Span length ranged from three to nine, with three trials per span length. More than one false response within the three trials of a span length ended the task. The length of the last span containing two out of three correct trials was extracted as the DS performance value.

Selective attention

To measure selective attention, a Victoria version of the Stroop Color-Word (VST[63]); was conducted. Participants were instructed to indicate the ink color of each word by pressing a corresponding color-coded response key, while ignoring the word's meaning. In the congruent condition, the word and its ink color matched (e.g., the word "blue" displayed in blue ink), creating no conflict between reading and color identification. In the incongruent condition, the word denoted a color different from its ink color (e.g., the word "red" displayed in green ink), requiring participants to inhibit the automatic reading response and instead respond based on ink color. In the neutral condition, non-color words (e.g., house, rock, tree) were presented in various ink colors, serving as a baseline without semantic interference. A total of 144 trials were equally divided into 48 neutral (W), congruent (C), and incongruent (CW) trials, respectively. Response timeout occurred after 3000 ms. The inter-stimulus interval was set to 1000 ms. A self-paced practice phase allowed participants to familiarize themselves with the color-to-response key mapping. The VST performance score was

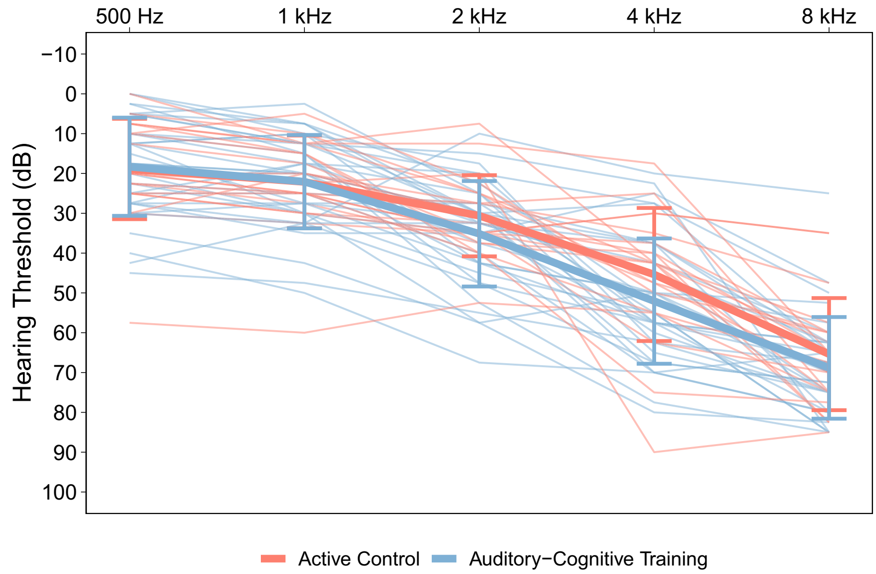

Fig. 7 | Pure-tone hearing thresholds over frequencies from 5 to 8 kHz. Note: Shown are individual pure-tone thresholds per frequency, ranging from 500 to 8000 Hz. Bolt lines represent pure-tone thresholds averaged across all members of the respective group. For participants using hearing aids, values reflect aided thresholds. Whiskers display the standard deviation of each group at each frequency. EG-HL = Experimental Group; CG-HL = Control Group.

calculated by taking the predicted number of correct responses and subtracting the total number of correct responses from it (Golden, 1978). The predicted number of correct trials is calculated using the count of correct item responses within 45 s for each condition, using formula (1). Subsequently, the total interference score is extracted using formula (2), where higher values indicate better inhibition performance.

$$Pcw = 45/((45 \times W) + (45 \times C))/(W \times C) \tag{1}$$

$$IG = CW - Pcw \tag{2}$$

Divided attention

Divided Attention was tested using a visual dual task paradigm (DT; Mueller & Piper[64],. Two arrows were presented, where the bottom arrow was always red, and the top arrow was always blue. Arrow directions varied with each trial, with participants indicating arrow directions to three different response conditions. Depending on the trial condition, participants were asked to indicate the direction of either just the blue or red arrow (single arrow trial), or both arrows simultaneously (dual arrow trial). Trials of each condition were presented in blocks of 25, in addition to 8 initial practice trials each time the response condition changed. Per round, one block of each response condition was presented. A complete DT measurement included four rounds, resulting in a total of 300 evaluated trials. Arrow directions were indicated using two keys (left, right) per arrow (blue, red). As a metric of performance, reaction time differences were calculated between single and dual arrow trials. The emerging dual-cost index was calculated as shown in formula (3), where DTC is the final dual-cost index and reaction times are taken from single and dual arrow trials.

$$\mathrm{DTC} = \frac{\mathrm{Reaction\_Time}_{dual} - \mathrm{Reaction\_Time}_{single}}{\mathrm{Reaction\_Time}_{single}} \tag{3}$$

Audiometry

To assess individual sensory hearing capacity, a pure-tone audiometry was conducted using an Affinity Compact audiometer (Interacoustics, Middelfart, Denmark). Tones were presented via DD450 circumaural headphones (Radioear, New Eagle, PA, USA). During assessment, participants were seated in an electrically shielded booth with anechoic padding. Pure-tone audiometry was conducted binaurally for frequencies of 0.125, 0.25, 0.5, 1, 2, 4, and 8 kHz. Subsequently, pure tone averages (PTA) were calculated across measured frequencies between 0.5 and 4 kHz across both ears ($M_{\mathrm{PTA}} = 34.94$, $\mathrm{Range}_{\mathrm{PTA4}} = 20.21–60.7$, $SD_{\mathrm{PTA}} = 9.5$). Target values of unaided PTA spanned from 20—60 dB HL, reflecting a mild to moderate degree of HL[58]. Groups did not significantly differ in PTA ($t\,(40.25) = -0.89$, $p = 0.379$). The distribution of aided (if applicable) pure-tone thresholds is plotted in Fig. 7.

Speech-in-noise comprehension

To measure SIN comprehension, a naturalistic speech-in-noise comprehension paradigm was employed. Full female-spoken conversational sentences were presented in alternating levels of stereoscopically recorded cafeteria noise[65], followed by a directly related comprehension question. Participants gave brief vocal responses to each comprehension question, with responses being recorded through a NT1-A microphone (Røde Microphones, Sydney, Australia). The task included three different noise conditions labeled as no-noise (SNR: not applicable), low-noise (SNR: 8 dB), and high-noise (SNR: 3 dB). Each noise condition contained 30 trials, with noise intensity always remaining stable for a block of five trials. Each session contained 90 trials with independent sentence sets for pre- and post-measurements, respectively. After the end of each block, participants were cued to indicate their perceived listening effort (LE) concerning the most recent trial block, using a 10-point Likert scale (NASA-TLX; Hart & Staveland[66],. Speech was continuously presented at 70 dB normalized sound pressure level (SPL) on a KEF Q150 loudspeaker (KEF, Maidstone, England) positioned at 0° azimuth. Stimulus loudness was controlled by an RME Fireface UCX II audio interface (RME, Haimhausen, Germany). Sentence length ranged from 5.0 to 8.0 s, while response timeout for comprehension questions occurred after 15 s. The inter-stimulus interval varied between 2.5 to 3.0 s, following a rectangular random distribution function. Accuracy scores were calculated as the percentage of correct responses per noise condition. Furthermore, accuracy was averaged across conditions to obtain the overall task accuracy score. An example of sentence-question pairing of the paradigm is shown in Table 1.

The stimulus set consisted of 180 sentences, with an average duration of 6.82 seconds. Sentences were spoken in standard German, with speech content covering ten conversational topics (History, Transportation, Geography, Culture, Traditions, Economy, Fauna, Flora, General trivia). Sentences were drawn from work of Frei and Giroud[67]. Noise snippets were extracted as 11 s-long snippets of stereoscopically recorded cafeteria noise from the HRIR database[65]. Noise snippets were fixedly assigned to specific sentences, forming fixed noise-sentence pairs. The order of presentation within each trial block was randomized. Both speech stimuli and noise files

Table 1 | Example set of speech-in-noise sentence, question, and possible response

Stimulus:	In 1933, a shy black panther escaped from the zoo and remained missing for a total of 10 weeks.
Question:	What escaped the zoo in 1933?
Response:	A shy black panther.

Note: The table illustrates a single trial from the speech-in-noise comprehension task. “Stimulus” is the sentence heard by the participant, “Question” is the comprehension prompt presented immediately afterward, and “Response” is the expected correct answer used for scoring.

were denoised and subsequently normalized so that the root mean square of audio levels equaled 70 dB SPL. Comprehension questions were designed so that there was only one correct answer. The position of target words within sentences varied across trials.

Speech-in-noise intelligibility

A commonly used clinical speech-in-noise perception task, the Oldenburger sentence test (OLSA;Kuehnel et al.,[68]), was employed to investigate speech-in-noise intelligibility. The test procedure aims to determine the 50% speech-perception threshold (SRT) from a total of 20 noise-embedded verbally spoken matrix sentences. Notably, each sentence adheres to a fixed *Name Verb Number Adjective Object* structure. Words for each sentence structure category are drawn at random—purposefully hindering contextual word inference. Sentences were always accompanied by speech-shaped white noise of varying loudness. Both noise and target sentence were presented at an initial 65 dB sound pressure level (SPL) and continuously adapted with each consecutive trial. Specifically, if less than three words were correctly reproduced, the SNR of the subsequent trial went down by 0.3 dB, increasing otherwise. This eventually results in the signal-to-noise ratio (SNR) at which 50% of a trial’s words can still be accurately reproduced.

Subjective hearing capacity

For the evaluation of subjective hearing capacity, a short version of the Speech, Spatial, and Qualities of Hearing Scale (SSQ-12; Noble et al.,[69]) was used. This scale aims to evaluate the subjective perception of hearing ability or disability in complex listening scenarios, such as the perception of speech in various competing sound contexts as well as spatial hearing attributes, such as sound direction, distance, and movement[70]. The SSQ-12 consists of twelve items, describing various complex daily life listening scenarios. For each listening scenario, participants are asked to indicate how well they would perform. Responses are given on a Likert scale, ranging from 0 (“No, not at all.”) to 10 (“Yes, perfectly."). Responses are averaged across all items to derive the SSQ total score.

Auditory-cognitive training

Participants in the experimental group were instructed to complete a 4-week long ACT program, training for at least 20 minutes a day, five days a week. ACT activity was logged using the Cognifit research platform (CogniFit Inc., San Francisco, USA). Training contents were drawn from the ActivEar mobile application (CogniFit[71]). The ActivEar training module is based on CogniFit’s validated cognitive training framework. ActivEar comprises a tailored subset of original CogniFit training games, supplemented by proprietary tasks specifically developed to target auditory-cognitive functions. Following initial installation of the training application, the CogniFit’s cognitive assessment battery CAB™ was conducted to obtain cognitive baseline performance. These initial performance values then informed the automated synthesis of a personalized training plan. Training plans always prioritized improvement of cognitive abilities the respective participant scored lowest on. Cognitive ability scores were updated after each training unit, ensuring the training plans were adapted continuously. Each training unit consisted of two training tasks and a concluding assessment task. While training tasks were generally more playful, assessment tasks were highly constrained and derived from established neuropsychological test paradigms. A list of assessment tasks, and classical paradigms that assessment tasks and hence on-training cognitive performance evaluation were modelled after, can be found in Supplementary Table 6.

Foreign language training

Active control participants underwent four weeks of foreign language training. The training was conducted in the computer-based language learning software Duolingo. Identical to participants undergoing ACT, active controls were asked to train for at least 20 min a day, five days a week. Notably, our choice of foreign language learning as an active control condition was further informed by prior work[72] successfully employing Duolingo as in active control during their auditory perception related training study. In their study, Schmitt and colleagues showed foreign language training to be cognitively engaging yet orthogonal to the auditory and attentional mechanisms targeted by the experimental training intervention[72].

Study procedure

Participants were randomly assigned to either the experimental group, which received four weeks of computer-based ACT, or an active control group, which engaged in four weeks of computer-based foreign language training. The study comprised three phases: pre-training, post-training, and a two-month follow-up. At the pre-training phase, testing was divided into two sessions. The first session included participant screening, pure-tone audiometry, the OLSA, and the full cognitive test battery. The second pre-training session consisted of the naturalistic speech-in-noise comprehension task, during which electroencephalographic (EEG) data were recorded; EEG measures were not analyzed for the present paper but are mentioned here for completeness. Following the second pre-training session, participants received in-person instructions and a tutorial for their assigned training program. Training was then completed at home over a four-week period, with participants instructed to train for at least 20 minutes per day, five days per week. Training activity in the ACT group was automatically logged by the application, while in the control group, it was recorded in paper-based learning diaries. On-demand technical support was available throughout the training phase, and participants were instructed not to undertake any other auditory or cognitive training during this period. Post-training testing followed the same two-session format as the pre-training phase. The first post-training session comprised pure-tone audiometry, the OLSA, and the cognitive test battery. The second post-training session involved the speech-in-noise comprehension task with concurrent EEG recording. At the two-month follow-up, only the first session was conducted, including hearing tests (pure-tone audiometry and OLSA) and the cognitive test battery.

All laboratory-based testing was conducted in an electrically and acoustically shielded booth. Hearing aid users wore their devices during all testing sessions, with individual device settings kept constant across the study. All participants provided written informed consent prior to participation and received monetary compensation. The study was approved by the local ethics committee (University of Zurich Ethics Commission, approval number 23.04.24) and conducted in accordance with the Declaration of Helsinki.

Group comparisons at baseline

Prior to elaborate statistical modelling and analysis, it was ensured experimental and control group displayed comparable baseline values for all measured sample characteristics. This served the purpose of identifying group differences, on variables of non-primary interest, that would potentially confound or drive training effect estimates. Thus, groups were compared regarding age, PTA, OLSA, subjective hearing ability (SSQ), global cognitive status (MoCA), working memory capacity, phonological short-term memory, divided attention, and selective attention (see Fig. 1). Variables that displayed significant differences between groups were included as covariates in subsequent modelling approaches. Group comparisons were conducted using Welch’s t-test, which is recommended when homogeneity of variance cannot be assumed and is robust to deviations from normality[73]. Given that Shapiro–Wilk tests indicated non-normal distributions for several baseline variables, Welch's test was chosen for its robustness to both unequal variances and non-normality in moderately sized samples. Reported p-values were Bonferroni-corrected for multiple comparisons, controlling for Type I error inflation. All steps of statistical analysis were conducted in R Studio (Version 2024.04.2[74]).

Effects regarding on-training and off-training cognitive performance

To evaluate the effects of ACT on cognitive performance measured by lab-based off-training and intervention app on-training tasks, Generalized Linear Mixed Models (GLMM) with Gaussian distributions and identity link functions were chosen. Outcome variables for both on- and off-training cognitive performance included phonological short-term memory, working memory, selective attention, and divided attention. Off-training task values were transformed to across-group percentile ranks to match on-training task performance metrics. All outcome measures were analyzed separately at the subject level. The fixed effects in each model included session (categorical; three levels: pre-training, post-training, follow-up) and group (categorical; two levels: CG-HL, EG-HL), as well as the interaction between these factors. Random slopes could not be estimated given that cognitive scores were comprised of a single average value per session, rather than item-level scores. A schematic model configuration for cognition can be seen in formula (x), with the exact configuration of the OLSA model in formula (4).

$$score \sim session * group + age_z + pta_{4000_z} + hearing\_aid + (1|subject) \tag{4}$$

Effects on speech-in-noise comprehension

Examining whether speech-in-noise (SIN) comprehension can be improved by means of auditory cognitive training (ACT) posed the primary research question of the present study. Given their conceptual relevance, age, pure-tone hearing thresholds, and hearing aid use were added as covariates to examine their potential effect on SIN comprehension performance and its training-related changes.

The analysis was conducted at the item level, with the dependent variable response coded as binary (1 = correct, 0 = incorrect). Given the hierarchical data structure, with repeated measures for each participant and items consistently presented within specific noise conditions, a GLMM with a binomial distribution and logistic link function was deemed appropriate.

Model configuration started with a maximal random effects structure justified by the study design[75], followed by iterative simplification of the model's random effects structure till a non-singular fit was achieved. The maximal model encompassed fixed effects for session (categorical; two levels: pre-training, post-training), group (categorical; two levels: CG-HL, EG-HL), hearing aid use (categorical; two levels: yes, no), alongside the covariates hearing loss (continuous; PTA4000) and age (continuous; in years). Continuous variables were z-standardized before model-inclusion. Additionally, random intercepts were included for both subjects and items, with item effects nested within noise levels to account for item-specific variability associated with the noise condition in which they were presented. To capture individual differences in noise susceptibility, random slopes for noise level were fitted by participant.

Contrasts were configured using sum-coding, so that lower-level effects are estimated at the level of the grand mean and interpreted accordingly[76].

Models were fitted using the *glmer* function from the lme4 package[77] in R. Fixed effects were interpreted using odds ratios (ORs) and 95% confidence intervals (CIs). Model performance was evaluated using marginal and conditional R^2[78]. Model assumptions were assessed using the *performance* package[79], and predicted probabilities were visualized using the *ggeffects* package[80].

Exact model configuration is shown in formula (5).

$$\begin{aligned} response \sim\ & session * group + noise_{level} + pta_{4000_z} + age_z \\ & + hearing\_aid + \left(1 + noise_{level} + session|subject\right) \\ & + (1 + noise\_level + session|noise\_level : item) \end{aligned} \tag{5}$$

Effects of auditory-cognitive training on subjective hearing and listening effort

Subsequent investigation focused on potential training effects concerning subjectively rated hearing performance on both the SSQ questionnaire and LE reported during SIN comprehension. Therefore, SSQ and LE values were extracted and analyzed for each session. Restricted value range (0–10) on SSQ and LE items was accounted for by means of fitting a GLMM with a logistic link function and beta distribution, rather than a Gaussian one. Prior to modelling SSQ and LE scores were divided by 10 to fit a binary value range for logistic regression. As before, model fitting was conducted using the glmmTMB package[81]. By-subject random intercepts were fitted to account for data dependencies. Again, random slopes were not estimated, given that SSQ and LE scores consisted of a single average value per session. A two-way interaction was fitted for session and group to evaluate whether training effects differed between treatments. As for factor-coding, sum-coding contrasts were applied once more. Notably, an identical model approach was chosen to evaluate training effects on OLSA performance, with the addition of PTA and age as covariates and a Gaussian rather than a binomial configuration. Schematic configuration of the model syntax can be seen in formula (6).

$$score \sim session * group + pta_{4000_z} + age_z + hearing\_aid + (1|subject) \tag{6}$$

Data availability

Utilized data and scripts of conducted analyses are made publicly available on OSF under https://osf.io/prmvc/?view_only=13709f0a94d748bf8570880111b8f1b7. No part of the study procedures and analyses was preregistered before the research was conducted.

Received: 12 June 2025; Accepted: 2 October 2025;
Published online: 19 November 2025

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Acknowledgements

We would like to thank Daniela Vrbova, Fabian Cabalzar, and Nadine Jait for their help on data acquisition and participant recruitment. Further, we would like to thank Sarah Baur and Patricia Reuter-Lorenz for their constructive feedback on multiple stages of the manuscript. Finally, gratitude is directed at Stefan Elmer and Vanessa Frei for their conceptual insights and suggestions on statistical modelling. The study was conducted at the Linguistics Research Infrastructure (LiRI) laboratory at the University of Zurich, with a particular acknowledgment of the technical support provided by Andrew Clark. We thank the University Research Priority Program "Dynamics of Healthy Aging" (UZH3) and the Healthy Longevity Center at the University of Zurich for their valuable support in participant recruitment. Lastly, we would like to thank the Swiss National Science Foundation (SNF) and Sonova AG for their financial support of this study.

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J.O.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Project administration, Writing—Original Draft, Visualization. N.G.: Conceptualization, Validation, Writing—Review and Editing, Supervision, Funding acquisition. S.S.: Conceptualization, Review & Editing, Supervision. S.M.: Conceptualization, Review & Editing, Supervision.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41539-025-00369-4.

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